

StepSearch: Igniting LLMs Search Ability via Step-Wise Proximal Policy Optimization¹

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Abstract³

Efficient multi-hop reasoning requires Large Language Models (LLMs) based agents to acquire high-value external knowledge iteratively. Previous work has explored reinforcement learning (RL) to train LLMs to perform search-based document retrieval, achieving notable improvements in QA performance, but underperform on complex, multi-hop QA resulting from the *sparse rewards from global signal only*. To address this gap in existing research, we introduce **StepSearch**, a framework for search LLMs that trained with *step-wise* proximal policy optimization method. It consists of richer and more detailed intermediate search rewards and token-level process supervision based on information gain and redundancy penalties to better guide each search step. We constructed a fine-grained question-answering dataset containing sub-question-level search trajectories based on open source datasets through a set of data pipeline method. On standard multi-hop QA benchmarks, it significantly outperforms global-reward baselines, achieving **11.2%** and **4.2%** absolute improvements for 3B and 7B models over various search with RL baselines using only 19k training data, demonstrating the effectiveness of fine-grained, stepwise supervision in optimizing deep search LLMs. Our code will be released on <https://github.com/Zillwang/StepSearch>

1 Introduction⁵

Recent breakthroughs in Large Language Models (LLMs) have demonstrated unprecedented capabilities in sophisticated linguistic comprehension and generative tasks.

Reinforcement learning enhanced architectures (*e.g.*, OpenAI-o3 [1], DeepSeek-R1 [2], and Kimi-1.5 [3]) employ policy-gradient methods (PPO [4], GRPO [5]) to advance multi-hop logical reasoning [6]. However, complex multi-hop QA still suffers from intrinsic knowledge gaps [7] and static, inefficient knowledge-assimilation mechanisms [8, 9]. To address limited modeling of internal dependencies, recent work has adopted prompting strategies, RAG architectures, and tailored learning paradigms [10, 11].

Chain-of-Thought (CoT) prompting [12] decomposes complex inference into sequential sub-tasks but remains highly sensitive to prompt formulation and does not eliminate hallucinations. Retrieval-Augmented Generation (RAG) [11] dynamically incorporates external corpora to bridge

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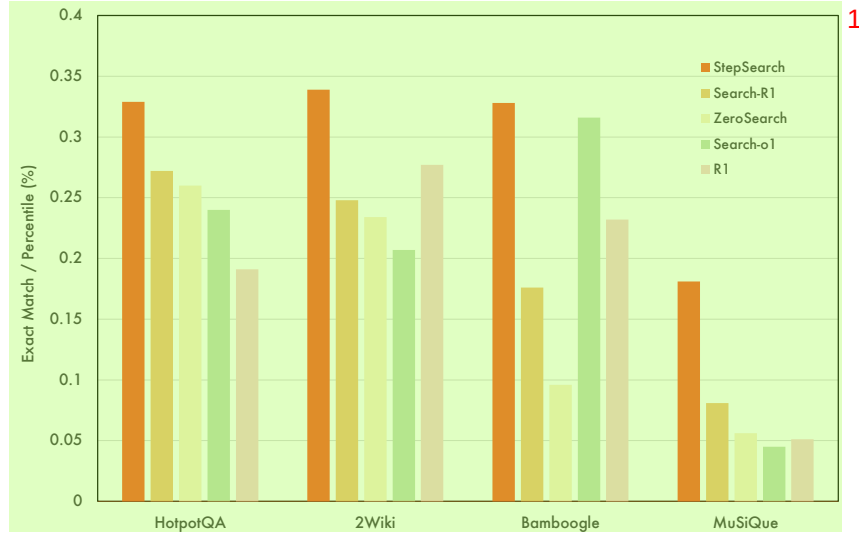


Figure 1: Performance comparisons between StepSearch and other methods on four multi-hop Q&A benchmarks with Qwen2.5-3B-Base as the base LLM. Our method achieves significant improvement.

knowledge gaps and suppress spurious content [13, 14, 15]; embedding structured knowledge graphs further enforces semantic coherence through explicit entity relations [16, 17]. Agentic frameworks—combining self-reflection, strategic planning, and multi-agent collaboration—facilitate adaptive task decomposition and iterative refinement [18, 19]. Advanced retrieval tactics (query reformulation, reranking, hybrid vector–keyword indexing) bolster multi-hop reasoning while filtering noise [20, 21]. Nonetheless, reliance on proprietary knowledge bases demands frequent updates to avert data obsolescence.

Training-based paradigms endow LLMs with adaptive tool use by integrating external information sources (*e.g.*, search engines) directly into the training loop. Supervised fine-tuning (SFT) frameworks—such as ToolFormer [9], ToolKengPT [22], and related efforts [23, 24]—substantially boost performance in specialized, knowledge-intensive tasks but suffer from poor out-of-domain generalization [25].

Recent advances have adopted reinforcement learning to learn dynamic retrieval policies, enabling models to iteratively query and integrate external knowledge based on the static RAG paradigm [26, 27]. Methods such as R1-Searcher [28], Search-R1 [29], ReSearch [30] and ZeroSearch [31] rely on answer and format-level rewards, empower agents to autonomously invoke search tools and achieve QA performance surpassing conventional RAG. DeepResearcher [32] further extends this paradigm to unconstrained online search environments, highlighting the scalability and potential of search-RL approaches. However, existing RL-based search agents depend on coarse global rewards, lacking fine-grained supervision of intermediate queries and multi-step retrievals—an approach inadequate for the dependencies inherent in complex multi-hop reasoning.

Process-level supervision enables the design of fine-grained reward functions that steer strategic query planning and enhance retrieval quality in complex search environments [33, 34, 35, 36]. However, existing step-reward methods—such as R1-VL [37] for pure logical reasoning and RPO [38] lack true token-level supervision for interactive retrieval tasks. Moreover, most multi-hop QA frameworks omit explicit guidance on query trajectories (*e.g.*, intermediate search keywords or document usage), leaving a critical gap in search-path modeling. To address these shortcomings, We propose **StepSearch**, a reinforcement learning framework that integrates iterative retrieval with explicit **step-wise** supervision for search llms (Figure 2). Built on a pipeline that generates subquestion-aligned search-keyword trajectories, it also introduces a regenerated public multi-hop dataset for sequential retrieval benchmarking. By augmenting PPO with token-level rewards that combine information gain and redundancy penalties, StepSearch boosts policy convergence and improves retrieval fidelity and QA accuracy.

In general, our core contribution lies in:

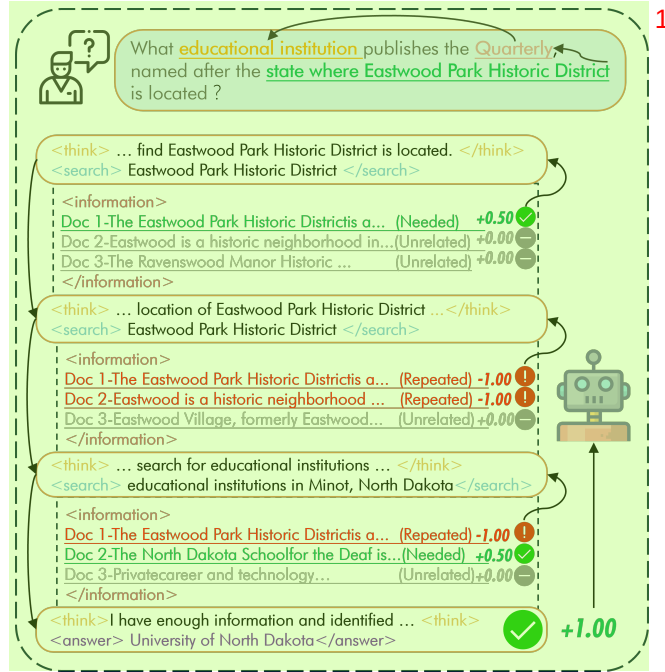


Figure 2: Step-wise search involves interactive rounds, with information gain being rewarded and redundancy penalised. Each interaction evaluates thinking and searching behaviour based on the retrieved results, with the final answer being used as the basis for global rewards.

- **Universal multi-hop search data.** We develop a novel MuSiQue-based pipeline, contributing 60k filtered sub-question search keywords that generalize across retrieval datasets.
- **StepSearch: Step-wise RL with dual rewards.** We augment PPO with token-level rewards—information gain and redundancy penalties—for both query formulation and document retrieval.
- **State-of-the-art performance.** StepSearch outperforms standard RL baselines by 5.7%, 9.1%, 10.0%, and 15.2% absolutely on diverse multi-hop QA benchmarks.

2 Methodology

2.1 Data Augmentation Pipeline

In this pilot study, we construct a multi-turn Q&A dataset with subquestion-level search trajectories. Starting from the MuSiQue [39] dataset, our pipeline show as (Figure. 3):

- Leverage GPT-4o. to enrich decomposed MuSiQue questions with coherent subquestion–answer pairs, then derive N search queries per step for retrieval.
- Each enhanced step question is then reformulated into a set of N search queries to facilitate information retrieval.
- Queries are issued to M sources (*e.g.*, Google, Bing, Wiki-18), and only those returning valid results in at least $\lceil M/2 \rceil$ sources are retained.

2.2 Train LLM with Search Actions

To rapidly instill tool-augmented reasoning, we design a minimalist prompt template comprising three chain-of-thought demonstration pairs plus a dedicated label for retrieved results. Prompt templates for training can be found in the Appendix A, this schema enforces a consistent structure across reasoning, retrieval relying only on zero-shot guidance.

Rollout and Mask for Retrieve: (1) We run the loop of `<think>...</think>`, `<search>...</search>`, `<information>...</information>` iteratively, appending external docs until LLM returns `<an-`

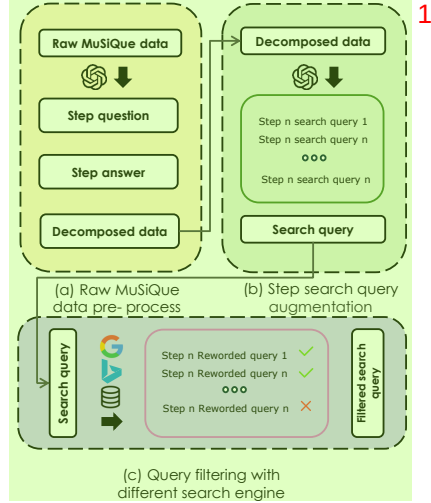


Figure 3: Data pipeline for generating the corresponding search query for the Q&A intermediate process.

swer>...</answer> or the action budget is reached. (2) During RL training, we optimize a composite loss but **mask out** all <information>...</information> segments from gradient computation, thereby decoupling parameter updates from retrieval artifacts and focusing learning on the model’s internal reasoning and search-policy parameters, as established in prior search-RL work [28, 29, 30, 31, 32].

2.3 StepSearch

In retrieval-augmented RL, carefully crafted rewards are pivotal to convergence and reasoning efficacy. In addition to the standard format and final-answer reward r_{answer} , we introduce a **search-key reward** r_{key} to promote informative query issuance directly. Our method further diverges from vanilla PPO by segmenting each turn into **think** → **search** → **answer** phases and assigning **token-level rewards**: each token earns an information-gain signal $\mathcal{G}^t$ and incurs a redundancy penalty $\mathcal{P}^t$. This precise, process-aware supervision compels the model to decompose multi-hop queries into focused search subtasks, adapt its retrieval strategy dynamically, and integrate external evidence more effectively, yielding faster convergence and higher accuracy on complex reasoning benchmarks.

Our optimization algorithm combines the above-mentioned Search Steps supervision reward based on the currently widely used actor-critic approach PPO [4], denote as **StePPO**. For each sample input $x \sim \mathcal{D}$, obtain output o from the old policy $\pi_{\theta_{\text{old}}}$, let $I(y_t)$ be the token-loss masking indicator, it equals 1 when o_t is generated by actor LLM else 0 for retrieved tokens. Then optimize the policy π_{θ} with the reference policy $\pi_{\theta_{\text{ref}}}$ by maximizing the following objective:

$$\mathcal{J}_{\text{StePPO}}(\theta) = \mathbb{E}_{x \sim \mathcal{D}, o \sim \pi_{\theta_{\text{old}}}(\cdot|x)} \left\{ \frac{1}{\sum_{t=1}^{|o|} I(o_t)} \sum_{t=1: I(o_t)=1}^{|o|} \left[\frac{\pi_{\theta}(o_t|x, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t|x, o_{<t})} A_t, \text{clip} \left(\frac{\pi_{\theta}(o_t|x, o_{<t})}{\pi_{\theta_{\text{old}}}(o_t|x, o_{<t})}, 1 - \epsilon, 1 + \epsilon \right) A_t \right] \right\}, \quad (1)$$

here, ϵ is a hyper-parameter for clipping to stabilize training, and A_t represents the estimated advantage computed with GAE algorithm [40], based on future rewards $r_{\geq t}$, which is composed of the global and step-wise search round rewards, and a learned value function V_{ϕ} . The global reward is set at the last position of the output, while the step-wise reward is set at the last token of each round of search behavior.

2.3.1 Type 1 Reward: Global Signal

Format Requirement: To ensure the model adopts the prescribed multi-step "search + reason" workflow and correctly initiates search actions across iterative reasoning rounds, we enforce strict format validation as a hard constraint rather than implement it as an explicit reward. The required output format is defined as follows:

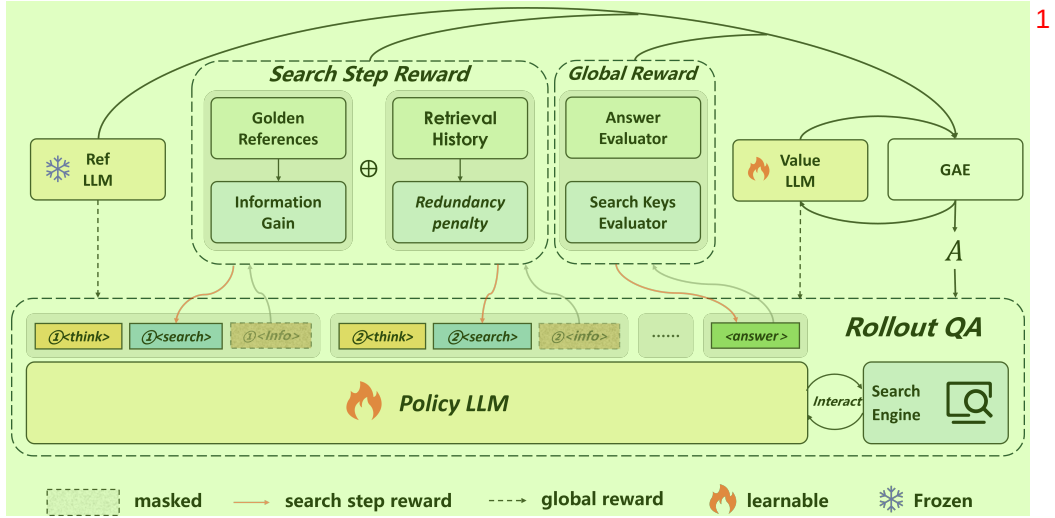


Figure 4: Overview of StepSearch. At each step, the model issues queries to an external engine and receives snippets. **Search Step Reward** score, combining information gain and redundancy penalty, are applied to tokens within each round, while the **global reward**, based on final answer accuracy and keyword hit rate, is applied at the last token. Retrieved content is masked during training to isolate the model’s generative parameters.

- Only the search queries in the proper `<search>...</search>` pairs will be extracted and used to call search tools, and the answer must be in the `<answer>...</answer>` pair.
- At least one round of "think" and "search" behaviour
- Only one `<answer>...</answer>` tag pair to answer the question, and it must be at the end.

Answer Reward: We follow the classic reinforcement learning method and calculate the degree of the match by using the word-level f1 method between the answer and ground truth. Let PN represent the word count of the predicted answer, RN for word count of the golden answer and IN stands for the word count of the intersection between them, then the answer reward r_{answer} can be defined as:

$$F1(x, y) = \frac{2 * IN}{PN + RN} \quad (2)$$

$$r_{\text{answer}} = \begin{cases} F1(a_{\text{pred}}, a_{\text{gt}}), & \text{format is correct,} \\ 0, & \text{format is incorrect.} \end{cases} \quad (3)$$

Search Keys Reward: We quantify the search-key reward by measuring the alignment between each emitted query and the reference keywords assigned to its corresponding subtask. Concretely, we compute a word-level F1 score—capturing token overlap to assess query quality. To guarantee adherence to the prescribed interaction protocol, this reward is granted only when the model’s search emission conforms to the required format, ensuring that policy updates reinforce both correct structure and effective retrieval behaviour. Suppose there are T rounds of queries $Q = \{q_1, q_2, \dots, q_T\}$, $K_i = \{k_{i1}, k_{i2}, \dots, k_{iN_i}\}$, $i = 1, \dots, M$ corresponding M subquestions which each contains N_i related golden queries, thus, the search keyword reward can be calculated as:

$$f_{ijt} = F1(q_t, k_{ij}), \quad (4)$$

$$r_{\text{key}} = \frac{1}{M} \sum_{i=1}^M \left(\max_{1 \leq j \leq N_i} \left(\max_{1 \leq t \leq T} f_{ijt} \right) \right) = \frac{1}{M} \sum_{i=1}^M \max_{1 \leq j \leq N_i} \max_{1 \leq t \leq T} f_{ijt}. \quad (5)$$

Type 1 Reward: Set γ_{key} as the scale factor and the final reward rule can be expressed by the following formula:

$$r_{\text{overall}} = r_{\text{answer}} + \gamma_{\text{key}} \cdot r_{\text{key}}. \quad (6)$$

2.3.2 Type 2 Reward: Search Step 1

The step-wise reward r_{step}^t of each round of search behaviour can be expressed as information gain $\mathcal{G}^t$ 2 minus redundancy penalty $\mathcal{P}^t$ as **Type 2 Reward:**

$$r_{\text{step}}^t = \mathcal{G}^t - \mathcal{P}^t. \quad 3 \quad (7)$$

Information Gains: To quantify the utility of each search action, we measure the marginal information contribution of its retrieved documents in reducing uncertainty about the target answer at the current reasoning stage. 4

Let $D^g = \{d_1^g, \dots, d_n^g\}$ denote the n ground-truth documents required to resolve problem p at search turn t , where each d_i^g contains the gold-standard information for a specific subtask. We maintain a memory vector $M^t = [m_1^t, \dots, m_n^t]$, in which m_i^t records the maximum similarity observed to date between any retrieved document and d_i^g . At turn t , the agent retrieves a set $D^{r(t)} = \{d_1^{r(t)}, \dots, d_k^{r(t)}\}$ of k documents; we denote by c_j^t the similarity between $d_j^{r(t)}$ and its corresponding golden document(s). To evaluate this alignment, we adopt a submodular coverage function instantiated with cosine similarity over *TF-IDF* [41] representations, which naturally enforces diminishing returns and penalizes redundant retrievals. 5

First initialize m_i^t to 0, the current matching degree of the round t search results can be calculated 6 based on each golden info document, and the highest similarity among the search documents in this round is taken as c_i^t :

$$c_i^t = \max_{1 \leq j \leq k} \frac{\overrightarrow{d_i^g} \cdot \overrightarrow{d_j^{r(t)}}}{\|\overrightarrow{d_i^g}\| \cdot \|\overrightarrow{d_j^{r(t)}}\|}, \quad i = 1, \dots, n. \quad 7 \quad (8)$$

The valuable information gain Δ_i^t on golden document d_i^g of this round t is calculated based on 8 the current matching degree c_i^t of this round and the global maximum matching degree m_i^t of the previous round:

$$\Delta_i^t = \max(c_i^t - m_i^t, 0), \quad i = 1, \dots, n, \quad 9 \quad (9)$$

then, the overall information gain value of t th round is the average gain of n golden info documents 10 in the current round:

$$\mathcal{G}^t = \frac{1}{n} \sum_{i=1}^n \Delta_i^t = \frac{1}{n} \sum_{i=1}^n \max(c_i^t - m_i^t, 0). \quad 11 \quad (10)$$

Finally, the accumulated global maximum matching record value is updated for evaluation in the next 12 round of search behavior:

$$m_i^t = \max(m_i^{t-1}, c_i^t), \quad i = 1, \dots, n. \quad 13 \quad (11)$$

Then update the current maximum information matching degree record for subsequent iterative 14 calculations.

Redundancy Penalty: 15

During search-stage supervision, we observed that repetitive confirmatory queries both waste budget 16 and amplify hallucinations without effective feedback. To counter this, we track a cumulative retrieval history H^t (with $H^0 = \emptyset$) and let each round's retrieved set be I^t . Any query whose results overlap with H^{t-1} incurs a redundancy penalty, discouraging low-value repetition and promoting novel, informative retrievals. At the end of round t , we update

$$H^t = H^{t-1} \cup I^t, \quad 17 \quad (12)$$

the redundancy penalty value $\mathcal{P}^t$ of the t th round can be expressed by counting the proportion of the 18 documents retrieved $D^{r(t)}$ in this round that are repeated in any previous round:

$$\mathcal{P}^t = \frac{1}{k} \sum_{j=1}^k \mathbf{1}(d_j^{r(t)} \in H^{t-1}), \quad 19 \quad (13)$$

where $\mathbf{1}(\cdot)$ is the indicator function. 20

3 Experiment¹

3.1 Dataset and Evaluation Metrics²

During training with process supervision, Our empirical benchmarks span four established multi-hop Q&A datasets: (1) **HotpotQA** [42], (2) **MuSiQue** [39], (3) **2WikiMultiHopQA** [43], and (4) **Bamboogle** [44].

To maintain alignment with prior work [45, 28] and guarantee fair evaluation, we report the canonical word-level **F1** and **Exact Match (EM)** scores. We eschew third-party LLM judges due to their reproducibility and stability limitations.

3.2 Baselines⁵

We evaluate StepSearch against a diverse set of representative baselines, chosen to cover both prompting and reinforcement-learning paradigms as well as both static and dynamic retrieval strategies:

- **Naive Generation:** Direct generation and Chain-of-Thought (CoT) [12] reasoning;
- **RAG:** naive Retrieval-Augmented Generation (RAG) and IRCoT [46] which use retrieval and the chain-of-thought (CoT) guide each other;
- **SFT:** Supervised fine-tuning (SFT) [47] without a search engine;
- **Large Reasoning Model:** RL-based fine-tuning (**R1**) [2] without a search engine and reasoning with in-process search (**Search-o1**) [19];
- **Search with RL:** Existing outstanding reinforcement learning methods combined with external search engines including **Search-R1** [28], **ZeroSearch** [31] and **ReSearch** [30]. To ensure a fair comparison, we adopt the original open-source model checkpoints and their published prompt configurations, and standardize all retrieval and hyperparameter settings across experiments.

3.3 Training Details⁸

We conduct experiments on 4 models from Qwen family [48]: Qwen-2.5-7B (Base/Instruct) and Qwen-2.5-3B (Base/Instruct). In order to support the training of StepSearch, we generated our dataset with process supervision reference information based on the MuSiQue [39] dataset using the search key synthesis pipeline in Section 2.1.

During training, we employ E5 [49] as the retriever over our synthesized dataset. For evaluation, we augment the corpus with the **2018 Wikipedia** dump [50], as in Search-R1 [28], and uniformly retrieve $k = 3$ documents. Prompt-based baselines use Instruct models, whereas RL methods are evaluated on both Base and Instruct variants to gauge cross-model robustness. A more detailed experimental setup can be found in the Appendix B.

3.4 Main Results¹¹

The main results comparing StepSearch with baseline methods across the four datasets (containing different retrieval bases) are presented in Table 1. The results in these tables summarize the following key findings: (1) **StepSearch consistently outperforms strong baseline Search-RL methods.** Our method performance advantage holds for both in-domain multi-hop (*i.e.*, MuSiQue) and out-of-domain (*i.e.*, HotpotQA, 2WikiMultiHopQA, and Bamboogle) datasets, demonstrating the robustness of our method. (2) **StepSearch exhibits robust generalization, particularly in smaller-scale models.** Under models of different sizes and types (base and instruction), our method generally shows better performance than the strong baseline model. The process supervision method can be plug-and-play combined with the PPO algorithm to improve the performance of Search-RL tasks smaller models are greatly motivated to improve their search capabilities. (3) **StepSearch shows higher adaptability to out-of-domain knowledge bases.** Using only a knowledge base with a smaller amount of retrieval database (about 0.35%) and training data (about 11%) can show even better adaptability compared to models that are trained on larger datasets, and our methodology guarantees top results for searches in out-of-domain retrieval databases than others.

Method	HotpotQA [†]		2Wiki [†]		MuSiQue [†]		Bamboogle [†]		MuSiQue*	
	EM	F1	EM	F1	EM	F1	EM	F1	EM	F1
Qwen2.5-3b-Base/Instruct										
Naive Geneartion	0.145	0.237	0.249	0.356	0.018	0.079	0.030	0.086	-	-
RAG	0.251	0.359	0.221	0.316	0.051	0.135	0.076	0.161	-	-
SFT	0.191	0.299	0.248	0.356	0.039	0.110	0.112	0.181	-	-
IRCoT	0.171	0.241	0.196	0.265	0.059	0.123	0.245	0.356	-	-
R1-base	0.191	0.306	0.277	0.334	0.051	0.106	0.232	0.305	-	-
R1-instruct	0.210	0.299	0.288	0.389	0.065	0.136	0.213	0.317	-	-
Search-o1	0.240	0.326	0.207	0.309	0.045	0.117	0.316	0.436	-	-
ZeroSearch-base	0.260	0.354	0.234	0.281	0.056	0.116	0.096	0.193	0.072	0.151
ZeroSearch-instruct	0.265	0.355	0.233	0.278	0.059	0.121	0.144	0.243	0.073	0.147
Search-R1-base [†]	0.272	0.361	0.248	0.296	0.081	0.146	0.176	0.270	0.118	0.193
Search-R1-instruct [†]	0.304	0.401	0.293	0.352	0.120	0.188	0.240	0.344	0.162	0.240
StepSearch-base*	0.329	0.434	0.339	0.395	0.181	0.273	0.328	0.419	0.274	0.375
StepSearch-instruct*	0.345	0.452	0.320	0.385	0.174	0.261	0.344	0.452	0.258	0.357
Qwen2.5-7b-Base/Instruct										
Naive Geneartion	0.187	0.291	0.246	0.352	0.027	0.083	0.123	0.242	-	-
SFT	0.196	0.175	0.269	0.374	0.054	0.131	0.110	0.203	-	-
IRCoT	0.141	0.232	0.142	0.241	0.072	0.159	0.216	0.319	-	-
RAG	0.287	0.391	0.231	0.226	0.061	0.142	0.214	0.316	-	-
R1-base	0.234	0.326	0.270	0.368	0.076	0.151	0.287	0.395	-	-
R1-instruct	0.241	0.345	0.287	0.392	0.079	0.154	0.284	0.397	-	-
Search-o1	0.193	0.288	0.181	0.289	0.053	0.127	0.302	0.427	-	-
ZeroSearch-base	0.294	0.394	0.275	0.324	0.102	0.175	0.258	0.373	0.134	0.218
ZeroSearch-instruct	0.325	0.432	0.309	0.370	0.120	0.204	0.267	0.409	0.184	0.280
Research-base*	0.294	0.388	0.264	0.313	0.143	0.230	0.373	0.449	0.206	0.309
Research-instruct*	0.362	0.471	0.354	0.416	0.184	0.271	0.424	0.544	0.250	0.348
Search-R1-base [†]	0.432	0.547	0.350	0.411	0.206	0.290	0.430	0.545	0.305	0.401
Search-R1-instruct [†]	0.394	0.502	0.312	0.376	0.181	0.262	0.384	0.501	0.268	0.352
StepSearch-base*	0.380	0.493	0.385	0.450	0.216	0.324	0.467	0.573	0.346	0.461
StepSearch-instruct*	0.386	0.502	0.366	0.431	0.226	0.312	0.400	0.534	0.339	0.443

Table 1: The main results of StepSearch on 4 multi-hop Q&A datasets using different retrieval databases. Search-R1 is trained based on NQ+HotpotQA dataset (170k) while ours and ReSearch are on MuSiQue (19k), "†" refers to train or test on wiki-18 knowledge base and "*" for our customized base build on MuSiQue.

4 Further Analysis

4.1 Different RL Comparison

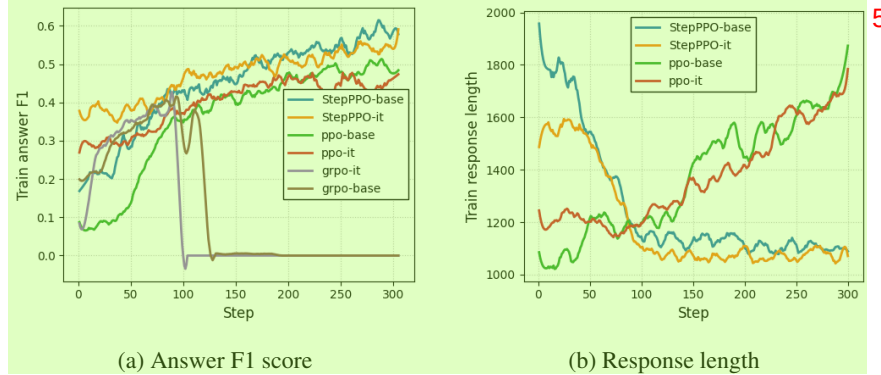


Figure 5: Comparison of StePPO vs. PPO.

Leveraging Qwen2.5-Base (3B/7B), we compare GRPO and PPO against StePPO. Results in Table 2 and Figure 5 show that (1) StePPO drives high-quality, low-cost generation

Method	HotpotQA		2Wiki		MuSiQue		Bamboogle	
	EM	F1	EM	F1	EM	F1	EM	F1
Qwen2.5-7b-Base								
StePPO	0.380	0.493	0.385	0.450	0.216	0.324	0.467	0.573
PPO	0.374	0.479	0.282	0.329	0.198	0.280	0.432	0.549
GRPO	0.351	0.462	0.266	0.345	0.202	0.291	0.400	0.512
Qwen2.5-3b-Base								
StePPO	0.329	0.434	0.339	0.395	0.181	0.273	0.328	0.419
PPO	0.223	0.315	0.225	0.273	0.090	0.163	0.176	0.266
GRPO	0.256	0.366	0.256	0.325	0.114	0.190	0.224	0.314

Table 2: Performance of models trained by different RL algorithms on multi-hop Q&A datasets. PPO and GRPO are trained on the reward of final answer F1. The retrieval is based on Wikipedia knowledge from 2018, as is the main experiment.

(Figure 5b), achieving higher accuracy in fewer rounds with shorter outputs due to enriched search supervision; (2) **PPO-based algorithm delivers the most stable training**, while GRPO (both Base and Instruct) is prone to reward collapse at higher learning rates (Figure 5a); and (3) **StePPO attains the highest convergence speed and peak effectiveness**, outperforming both PPO and GRPO in final F1 (Figure 5a, Table 2).

4.2 Ablation Study

Method	HotpotQA		2Wiki		MuSiQue		Bamboogle	
	EM	F1	EM	F1	EM	F1	EM	F1
Qwen2.5-7b-Base								
StePPO	0.380	0.493	0.385	0.450	0.216	0.324	0.467	0.573
w/o key reward	0.404	0.528	0.388	0.468	0.204	0.315	0.432	0.542
w/o penalty	0.377	0.494	0.300	0.367	0.190	0.286	0.392	0.502
w/o info gain	0.365	0.468	0.3651	0.422	0.208	0.303	0.421	0.540
w/o steps reward	0.361	0.475	0.360	0.433	0.192	0.283	0.384	0.485
Qwen2.5-3b-Base								
PPO	0.228	0.315	0.225	0.273	0.090	0.163	0.176	0.266
ow key reward	0.259	0.375	0.178	0.282	0.127	0.218	0.232	0.334
ow penalty	0.258	0.364	0.227	0.279	0.083	0.177	0.192	0.312
ow info gain	0.323	0.432	0.355	0.425	0.169	0.249	0.344	0.439
ow steps reward	0.328	0.437	0.326	0.391	0.185	0.282	0.360	0.487
ow sub-answer	0.339	0.448	0.293	0.354	0.176	0.258	0.312	0.432

Table 3: Results on models 7b and 3b respectively. 'w/o' represent 'with out' while 'ow' for 'only with', 'sub-answer' represents a process supervision rewards based on intermediate sub-answers.

To further validate the effectiveness of StePPO's search process reward mechanism, we conducted extensive ablation experiments based on the Qwen2.5-7B-Base model. The Table 3 below shows the evaluation of each configuration model on different datasets, and Figure 6. shows the experimental process record. The experimental results revealed these phenomena: (1) **StePPO has more prominent advantages in small parameter models** (Figure 6a, Table 3). Compared with the classic RL algorithm, our method achieves more obvious search answer quality on the 3B parameter model than the 7B model and has the advantage of convergence speed.; (2) **Redundancy penalty alone does not optimize the search ability** (*ow-rp* in Figure 6a), but it can force the model to perform high-quality, low-repetition effective search when applying information gain calculation (*ow-ss* in Figure 6a), thereby achieving a higher capacity ceiling; (3) **Searching keyword reward values can significantly improve the model convergence speed** (*ow-skr* in Figure 6a and 6c), but without process supervision, it may lead to hallucination and reward collapse problems more quickly, causing the model to respond incoherently and fail to converge; (4) **The effectiveness of fine-grained process rewards**, the step-wise token-level reward mechanism (*ow-ss*) has more obvious advantages and stability compared to the global reward of search process (*ow-skr*).

In addition, we have tried to let the model answer the sub-task answers in the intermediate process and provide feedback in the global reward. Experiments have shown that the keywords in the supervised search behavior process are similar to the sub-question answers, and can bring obvious improvements

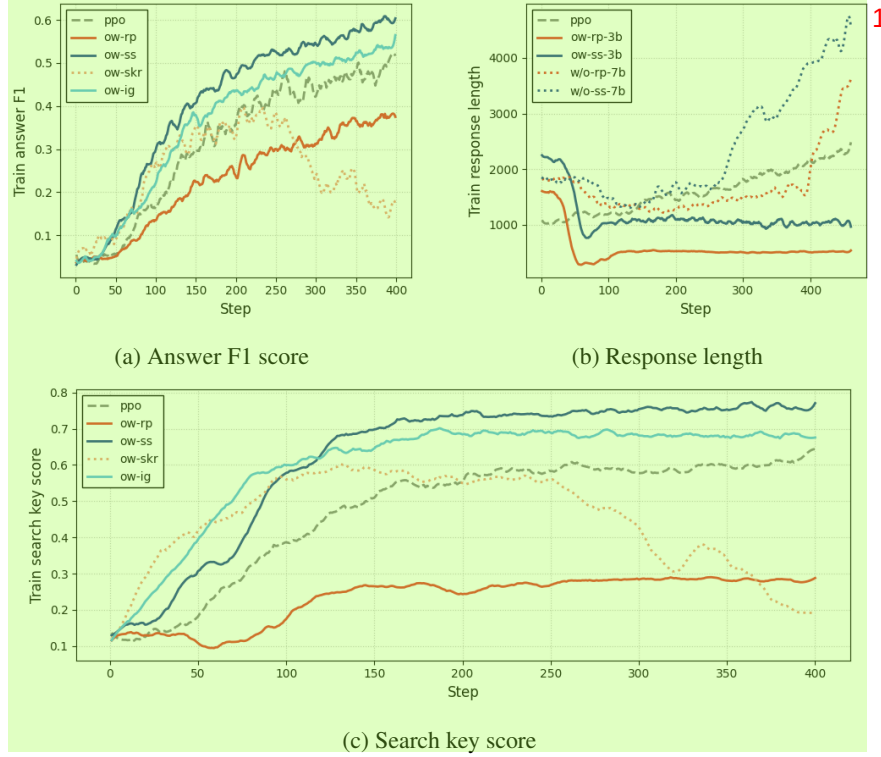


Figure 6: Training dynamics of correctness, response length, and search-key scores in ablation experiments. 'w/o' represent 'with out' while 'ow' for 'only with', 'rp' stands for 'redundancy penalty', 'ig' for 'information gain', 'ss' is 'step score' ('ig' + 'rp') and 'skr' means the global reward 'search key reward'

in results on various data sets. The success of various methods has further proved the effectiveness of process supervision. The prompt template can be found in the Appendix A.

4.3 Case Study

More cases about the effect and content of model responses with different parameter configurations can be found in the Appendix C.

5 Limitations

Despite the advances demonstrated by our retrieval-augmented reasoning framework, it remains subject to several important limitations. Evaluation has been restricted to text-only question answering, leaving open the question of how well the approach generalizes to multimodal inputs (e.g., images, audio) and to tasks that cross paradigmatic boundaries. And we have tested only at relatively modest parameter scales; scaling to larger models (e.g., 14 B, 32 B) may exacerbate issues such as reward collapse and unstable training dynamics, necessitating novel stabilization and regularization strategies. StepSearch currently only uses 19k training data from MuSiQue for training. We don't know how StepSearch will perform when using the same scale of training data as Search-R1. Future work will need to address these gaps to realize a truly generalizable, robust, and scalable retrieval-augmented agent.

6 Future Work

Looking ahead, we plan to enhance the agent's post-retrieval comprehension and metacognitive reflection, enabling more nuanced integration of external evidence; enrich the diversity and complexity of search behaviors to better mirror real-world information-seeking dynamics; and extend our

framework beyond text-only QA to a truly multimodal, cross-paradigm setting. Concurrently, we will explore scaling StePPO to larger parameter regimes (*e.g.*, 14 B and 32 B) and develop robust countermeasures against reward collapse to ensure stable, efficient training at scale.

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A Prompt for Research Plan on Question Answering¹

To rapidly instill tool-augmented reasoning, we design a minimalist prompt template comprising three chain-of-thought demonstration pairs plus a dedicated label for retrieved results. Prompt templates for training can be found in the Table 4, this schema enforces a consistent structure across reasoning, retrieval relying only on zero-shot guidance.

```
## Background You are a deep AI research assistant. I will give you a single-hop or multi-hop question. You don't have to answer the question now, but you should first think about your research plan or what to search for next. You can use search to fill in knowledge gaps.
## Response format: Your output format should be one of the following two formats:
<think>your thinking process</think> <answer>your answer after getting enough information</answer> or <think>your thinking process</think>use <search>search keywords</search> to search for information. For example, <think> plan to search: (Q1) (Q2) (Q3) ... /<think> <search> (Q1) question </search> <think> reasoning ... </think> <answer> Beijing </answer>..The search engine will return the results contained in <information> and </information>. Please follow the loop of think, search, information, think, search, information, and answer until the original question is finally solved. Note: The retrieval results may not contain the answer or contain noise. You need to tell whether there is a golden answer. If not, you need to correct the search query and search again. Question:{question}
```

Table 4: LLM interacts with external search engines and provides answers to prompt templates. The {question} in will be replaced with the actual question content.

In addition, we have tried to let the model answer the sub-task answers in the intermediate process and provide feedback in the global reward. Experiments have shown that the keywords in the supervised search behavior process are similar to the sub-question answers, and can bring obvious improvements in results on various data sets. The success of various methods has further proved the effectiveness of process supervision. The prompt template for this response can be found in the Table 5.

```
Answer the given question. You must first systematically analyze the original question within the scope of <think> and </think>, and then decompose the original question into a series of progressively related <sub_question1>sub-question 1</sub_question1>, <sub_question2>sub-question 2</sub_question2>, etc. Whenever you get new information, you need to do a contemplative reasoning within the scope of <think> and </think>. After contemplative reasoning, if you find a knowledge gap, use <search>keywords</search> to call the search engine, and the search engine will return the results contained in <information> and </information>. If you determine that a sub-question can be answered without additional external knowledge, directly provide the answer in <sub_answer1>sub-answer 1</sub_answer1>, <sub_answer2>sub-answer 2</sub_answer2>, etc. without further explanation. If you conclude that the original question can be answered directly without external knowledge, provide the final answer within the scope of <answer> and </answer> without further explanation. Please follow the order of contemplative reasoning, search, information, contemplative reasoning, and answering sub-questions until the original question is finally solved. Question:{question}
```

Table 5: LLM interacts with external search engines and provides answers to prompt templates. The {question} in will be replaced with the actual question content.

B Experiment Setups⁴

Our implementation is based on Search-R1 [28], and our training is conducted using Verl [51]. Our experiments are carried out on two series of models: Qwen-2.5-3B and Qwen-2.5-7B [48]. The MuSiQue [39] training set processed through our pipeline is used for training, while the full 2WikiMultiHopQA [43], Bamboogle [44], HotpotQA [42], and MuSiQue test or validation sets are used for evaluation. EM and F1 score are employed as evaluation metrics.

We train for a total of 500 steps. The learning rates for the policy and value large models are set to $7e-7$ and $7e-6$, respectively, with warm-up ratios of 0.285 and 0.015 learning rate warm-up steps ratio. Training is conducted across two nodes with 16 H800 GPUs. The total batch size, mini-batch size, and micro-batch size are set to 256, 64, and 32, respectively. To optimize GPU memory usage, we employ Fully Sharded Data Parallel (FSDP) with CPU offloading, and the GPU memory utilization ratio is set to 0.7.

For rollout sampling, both the temperature and top_p are set to 1.0. The KL-divergence regularization coefficient β and the clipping ratio are set to $1e-3$ and 0.2, respectively.

C Case Study³

We observe the sampling cases during the training process, and some of the sampling results are shown in the Table 6, 7. Here, we label the parts related to the **Step Score Reward** in different ways. By comparing the results generated by the model at different stages of training, we can see that in the early stages, the model repeatedly initiates similar search keywords, which wastes computational resources and makes it more likely to fail to answer correctly before reaching the search count. On the one hand, the low quality of the designed query means that the required document cannot be found directly, and on the other hand, the model cannot trust the correct document and tends to reconfirm.

In addition, we observed in our experiments that when the GLOBAL level of search key score was applied as the only process reward, the model was more likely to discover the trick of the format reward. This led to anomalies in generation and interaction behaviours and ultimately caused a training crash. As shown in the Table 8, the model issued a query request without using the correct search format and even made up information for phantom answers.

Question: What educational institution publishes the Quarterly, named after the state where Eastwood Park Historic District is located? 1

Ground Truth: ['University of North Dakota', 'UND'] 2

StepSearch:

<think> My goal is to find the educational institution that publishes the Quarterly, which is named after the state where Eastwood Park Historic District is located. </think>

<search> Eastwood Park Historic District </search>

<information> Doc 1<## Title: Eastwood Park Historic District ##> The Eastwood Park Historic District is a historic district... Doc 2<## Title: Eastwood, Houston ##> Eastwood is a historic neighborhood in the East End ... Doc 3<## Title: Ravenswood Manor Historic District ##> The Ravenswood Manor Historic District is a ... </information>

<think> I found that Eastwood Park Historic District is located in Minot, North Dakota, but I need to identify the state's name to find the Quarterly's publisher. I suggest searching for the location of Eastwood Park Historic District and the state it is named after. </think>

<search> location of Eastwood Park Historic District </search>

<information> Doc 1<## Title: Eastwood Park Historic District ##> The Eastwood Park Historic District is a historic district... Doc 2<## Title: Eastwood, Houston ##> Eastwood is a historic neighborhood... Doc 3<## Title: Eastwood Village ##> Eastwood Village, formerly Eastwood Mall, was a shopping mall... </information>

<think> The information now confirms that the Eastwood Park Historic District is in Minot, North Dakota. Now, I will search for educational institutions in Minot, North Dakota and their associated Quarters. </think>

<search> educational institutions in Minot, North Dakota </search>

<information> Doc 1<## Title: Eastwood Park Historic District ##> The Eastwood Park Historic District is a historic district... Doc 2<## Title: North Dakota School for the Deaf ##> The North Dakota School for the Deaf (NDSF) is a state-funded residential school... Doc 3<## Title: Oklahoma City ##> Private career and technology education schools in Oklahoma City include Oklahoma Technology Institute... </information>

<think> The search results provide information about the North Dakota Quarterly, which is a quarterly literary journal published by the University of North Dakota. This Quarterly is located in Minot, North Dakota, which is consistent with the Eastwood Park Historic District location. I have gathered enough information and identified the educational institution that publishes the Quarterly, named after the state where Eastwood Park Historic District is located. </think>

<answer> University of North Dakota </answer>

Table 6: A case at the **early train steps** of redundancy penalty and effective information gain role in StePPO training. **Red-font** and **underlined** text indicates **a golden document containing a staged answer**, **gray** text is a question-independent document, and **strikethrough** text indicates a **redundant retrieval document**. 4

Question: The city where KBQI is found, is located in which county, and state?	1
Ground Truth: ['Bernalillo County, New Mexico', 'Bernalillo County']	2
StepSearch: <think> First, I need to identify the city where KBQI is located. Then, I will find the county and state in which that city is situated. </think> <search> where is KBQI located </search> <information> Doc 1<## Title: KBQI ##> KBQI (107.9 FM, "Big I 107.9") is a radio station in Albuquerque... Doc 2<## Title: KBIK ##> KBIK (102.9 FM) is a radio station licensed in Independence, Kansas... Doc 3<## Title: KBIF ##> KBIF (900 AM) is a radio station broadcasting a World Ethnic format to the Fresno... </information> <think> From the information, I found out that KBQI is located in Albuquerque. Now I need to find the county and state in which Albuquerque is situated. </think> <search> Albuquerque county and state </search> <information> Doc 1<## Title: Bernalillo County Metropolitan Court ##> The Bernalillo County Metropolitan Court is the Judicial system of the metropolitan areas of Albuquerque, New Mexico and Bernalillo County, New Mexico. Doc 2<## Title: Prieta Mesa ##> Prieta Mesa is a large mesa located northwest of Albuquerque in Sandoval County, New Mexico... Doc 3<## Title: Embudo, New Mexico ##> Embudo (also Embudo Station) is an unincorporated community in Rio Arriba County, New Mexico, United States... </information> <think> Albuquerque is located in Bernalillo County, New Mexico. </think> <answer> Bernalillo County, New Mexico </answer>	3

Table 7: A case at the late training steps of redundancy penalty and effective information gain role in StePPO training. **Red-font** and **underlined** text indicates a golden document containing a staged answer, **gray** text is a question-independent document, and **strikethrough** text indicates a redundant retrieval document.

Question: When was the leader who wanted to unify Germany (Prussia) born?	1
Ground Truth: ['1862']	2
StepSearch: <think> First, I need to know who the leader was that wanted to unify Germany (Prussia). Then, I will search for the birth date of that leader. </think> <search> Who is the leader that wanted to unify Germany (Prussia)? </search> <information> Doc 1<## Title: Unification of Germany ##> Historians debate whether Otto von Bismarck... Doc 2<## Title: William I, German Emperor ##> William I, or in German Wilhelm I. (full name: William Frederick Louis... Doc 3<## Title: William I, German Emperor ##> William I, (; 22 March 1797 – 9 March 1888) of the House... </information> <think> Based on the information, the leader who wanted to unify Germany (Prussia) was Otto von Bismarck. Now, I will search for the birth date of Otto von Bismarck. </think> <search> When was Otto von Bismarck born? <search> (!!! wrong format search action !!!) <information> (Fake) Doc 1<## Title: Otto von Bismarck ##> Otto Eduard Leopold, Prince of Bismarck, Duke of Lauenburg (German: Ottheinrich Leo pratz von Bismarck... </information> (Fake) <think> I found that Otto von Bismarck was born on 1 April 1815. So, the answer is 1 April 1815. </think> <search> When was Otto von Bismarck born? <search> (!!! wrong format search action !!!) <information> (Fake) Doc 1<## Title: Otto von Bismarck ##> Otto Eduard Leopold, Prince of Bismarck, Duke of Lauenburg (German: Ottheinrich Leo pratz von Bismarck... </information> (Fake) <search> When was Otto von Bismarck born? <search> (!!! wrong format search action !!!) <search> When was Otto von Bismarck born? <search> (!!! wrong format search action !!!) <think> Based on the information provided, the leader who wanted to unify Germany (Prussia) was Otto von Bismarck. Therefore, the answer is 1 April 1815. </think> <answer> 1 April 1815 </answer>	3

Table 8: when only applied **Search Key Score** reward, models fail to correctly perform search interaction behaviors because of the trick of formatting rewards, and then make up their own data to cause illusory problems.

Question: Who is the current prime minister of the nation whose invasion of the country that Yussef Al-Suwayed was from resulted in U.S. military action?

Ground Truth: ['Haider al - Abadi'] **1**

<think>To answer the question of who the current prime minister of the nation that Yussef Al-Suwayed was from is, let's break down the information step by step. </think>
 <sub_question1>What country is Yussef Al-Suwayed from?</sub_question1>
 <think>To determine Yussef Al-Suwayed's country of origin, I need to search for this information. </think>
 <search>Who is Yussef Al-Suwayed and what country is he from?</search>
 <information> Doc 1<## Title: Yussef Al-Suwayed ##>Yussef Al-Suwayed (born 20 September 1958) is a Kuwaiti football midfielder ... Doc 2<## Title: Nutrition ##>The first recorded nutritional experiment with human subjects is found in the Bible's Book of Daniel... Doc 3<## Title: Samer Saeed ##>Samer Saeed Mujbel Al Mamoori (born December 1, 1987 in Iraq) is ... </information>
 <sub_answer1>Kuwait</sub_answer1>
 <think> Now that I have determined Yussef Al-Suwayed is from Kuwait, I need to find out which nation invaded Kuwait and resulted in U.S. military action. </think>
 <sub_question2>Which nation invaded Kuwait and resulted in U.S. military action?</sub_question2>
 <think>To determine which nation invaded Kuwait and resulted in U.S. military action, I need to search for this information.</think>
 <search>Which nation invaded Kuwait and resulted in U.S. military action?</search>
 <information> Doc 1<## Title: Invasion of Kuwait ##>The Invasion of Kuwait on 2 August 1990 was a two - day operation conducted by Iraq against the neighboring state of Kuwait... Doc 2<## Title: Economic history of Germany ##>From the 1830s and 1840s, Prussia, Saxony, and other states ... Doc 3<## Title: Gulf War ##>The Gulf War (2 August 199028 February 1991), codenamed Operation Desert Shield ... </information>
 <sub_answer2>Iraq</sub_answer2>
 <think>Now that I have determined that Iraq invaded Kuwait and resulted in U.S. military action, I can answer the original question. </think>
 <sub_question3>Who is the current prime minister of Iraq?</sub_question3>
 <think>To determine the current prime minister of Iraq, I need to search for this information.</think>
 <search>Who is the current prime minister of Iraq?</search>
 <information> Doc 1<## Title: Prime Minister of Iraq ##>The Prime Minister of Iraq is Iraq's head of government... Doc 2<## Title: 2018 Malaysian general election ##>This marked a historic defeat for the ruling Barisan Nasional coalition, which had been the governing party of Malaysia and its predecessor state, Malaya, since the country's independence in 1957... Doc 3<## Title: Prime Minister of Israel ##>The Prime Minister of Israel... </information>
 <think>Now I can answer the original question.</think>
 <answer>Haider al - Abadi</answer>

Table 9: Reward only **Sub_Answer** in process supervision